

Multiplication of integers



1

- a True b False

2

- a Negative b Positive c Positive d Negative

3

- a -24 b -24 c 24

4

- a -323 b -323 c 323

5

- a -1147 b -1147 c 1147

6

- a -30 b 30 c 30 d -30

7

- | | | | |
|--------|-------|--------|--------|
| a -16 | b 16 | c -64 | d 144 |
| e -144 | f 512 | g -216 | h -36 |
| i -45 | j 120 | k -9 | l 0 |
| m -44 | n 44 | o 16 | p -64 |
| q -140 | r 36 | s -144 | t -720 |

8

- a $r = 6$ b $n = 12$ c $q = -4$ d $t = -6$

9

- | | |
|--------------------------|--------------------------|
| a $-11 \times (-5) = 55$ | b $-6 \times 11 = -66$ |
| c $-7 \times 8 = -56$ | d $7 \times (-12) = -84$ |

10

- a -7 b 4

11 When we square a number the result is always positive. As -1 is negative there is no number x which we can square and get -1 .



Division of integers

12 Negative

- 13 a Negative b Positive c Negative d Positive

- 14 a -6 b -6 c 6

- 15 a -23 b 23 c 23

- 16 a -33 b -33 c 33

- 17 a 1 b -2 c -4 d 0
e -3 f -5 g -3 h -2
i 5 j 8 k -1 l 1
m -10 n -6 o -6 p 4

- 18 a $n = 9$ b $p = -88$ c $t = -8$ d $k = -48$

- 19 a $54 \div (-6) = -9$ b $-35 \div 5 = -7$
c $-55 \div 5 = -11$ d $(-72) \div (-8) = 9$

- 20 a -27 b 20

- 21 a 6 b 105 c -16 d -72
e 8 f -30 g 3 h -7

Applications

22 a Quadrants I and III



b Quadrants III and IV

23 a 12

b -5

c 11

24 a \$13 200

b \$2400

c \$10 800